

Sort directories by disk usage and display the largest ones

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Displays the top 10 directories with the largest disk usage in the current directory, sorted in descending order by size in human-readable format.

Language

bash

Category

disk-usage

Command

```
du -sh */ | sort -hr | head -10
```

Explanation

The command uses 'du -sh' to calculate the total disk usage for each directory in human-readable format. The '*/' glob pattern matches only directories (not files). The output is piped to 'sort -hr' to sort in descending order using human-readable number comparison, with the largest directories appearing first. Finally, 'head -10' limits output to the top 10 directories.

Tags

du, disk-usage, directories, storage, sort, system-administration

Dependencies

coreutils

Arguments

1. **PATH** (Optional): Directory path to analyze (default: current directory)
Default: .

2. **COUNT** (Optional): Number of largest directories to display (default: 10)
Default: 10

Examples

1. `du -sh */ | sort -hr | head -10` - Show top 10 largest subdirectories in current directory
2. `du -sh */ | sort -hr` - Show all subdirectories sorted by size without limit
3. `du -sh */ | sort -hr | head -5` - Show top 5 largest subdirectories
4. `du -sh */ | sort -hr | head -10` - Include hidden files and directories (those starting with dot)
5. `cd /var && du -sh */ | sort -hr | head -10` - Find largest subdirectories in /var

Output

```
847G    cache/
512G    data/
256G    backups/
128G    logs/
96G     documents/
64G     downloads/
32G     pictures/
16G     videos/
8.5G    archives/
2.1G    temp/
```

Notes

- The `'*/'` pattern matches only directories; use `'**'` to include files as well
- The `-h` flag makes output human-readable (K, M, G, T); use `-B` for a specific block size
- The `-s` flag summarizes each directory's total size instead of listing contents recursively
- Hidden directories (starting with `.`) are excluded; use `'./*'` or `'**'` (with `shopt`) to include them

- On systems with many subdirectories, results are calculated quickly since `du` only traverses each directory once
- Symbolic links are not followed by default; add `-L` flag to follow symlinks
- Results may vary if some directories are mounted on different filesystems or are inaccessible due to permissions

Warnings

- Results may be incomplete or misleading if you lack read permissions on some directories
- On network filesystems (NFS, SMB), the command may be slower due to network latency
- If dealing with millions of directories, the command can consume considerable memory and time
- Sparse files may cause inaccurate size reporting on some filesystems

See Also

- `find-largest-storage-users`
- `find-large-files-recursive`

Status

reviewed

Safety

safe

Shell

bash

Platforms

linux, gnu-linux, freebsd, openbsd, netbsd

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